Krishi Connect

2024-25 · water access, bamboo and community livelihoods

A social-research project connecting village water constraints, bamboo-based fog collection, drip irrigation and livelihood planning through questionnaires, interviews, models and community awareness activities.

Architecture



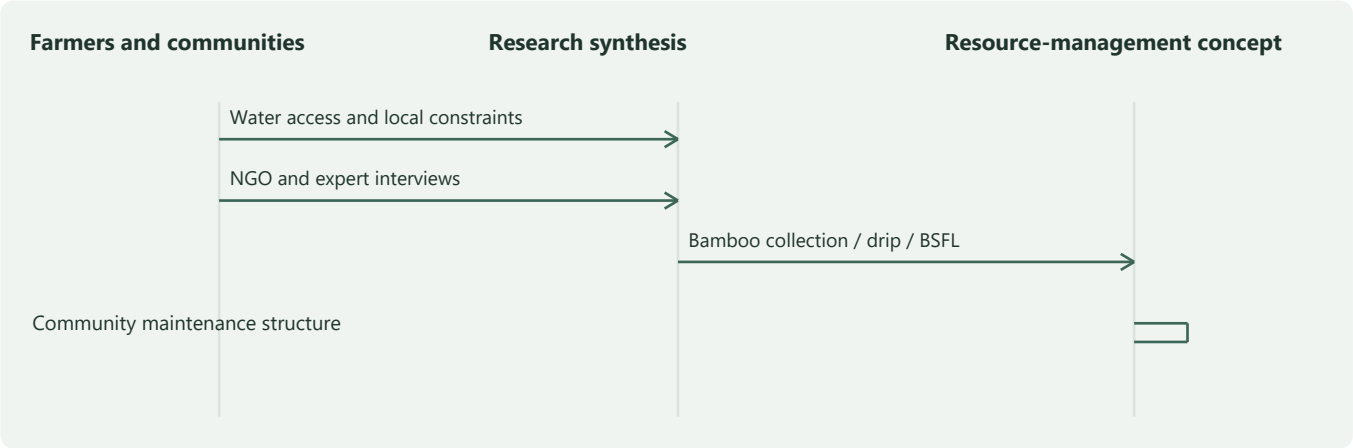
Component overview derived from the supplied source; version-specific details may differ.

Component	Implementation	Role
Community inquiry	Questionnaires and interviews	Understand local constraints
Research synthesis	Expert and institutional consultation	Evaluate proposed responses
Intervention design	Livelihood and awareness activities	Link technical options to community needs
Evaluation	Source-reported findings and limitations	Separate proposals from established outcomes

Community research project

Community research and design brief

Krishi Connect is the title on the 2024-25 portfolio cover, while its approval sheet and conclusion use Droplets of Hope. The project links rural water access, farmer livelihoods and tribal employment, drawing on village questionnaires, interviews and an explanatory model.



Research process

Questionnaires concern Tillore Khurd and Chikli, Ghosikheda and Pathan Piplya. The document also records a telephone interview with Dr Kamlesh Kumar of Samdarshi Seva Kendra and a farming/irrigation discussion with Dr Sumer Singh. Visits identify seasonal shortages, limited technology access and difficulties navigating schemes. The narrative supplies local context, not a random-sample estimate.

From concern to requirement

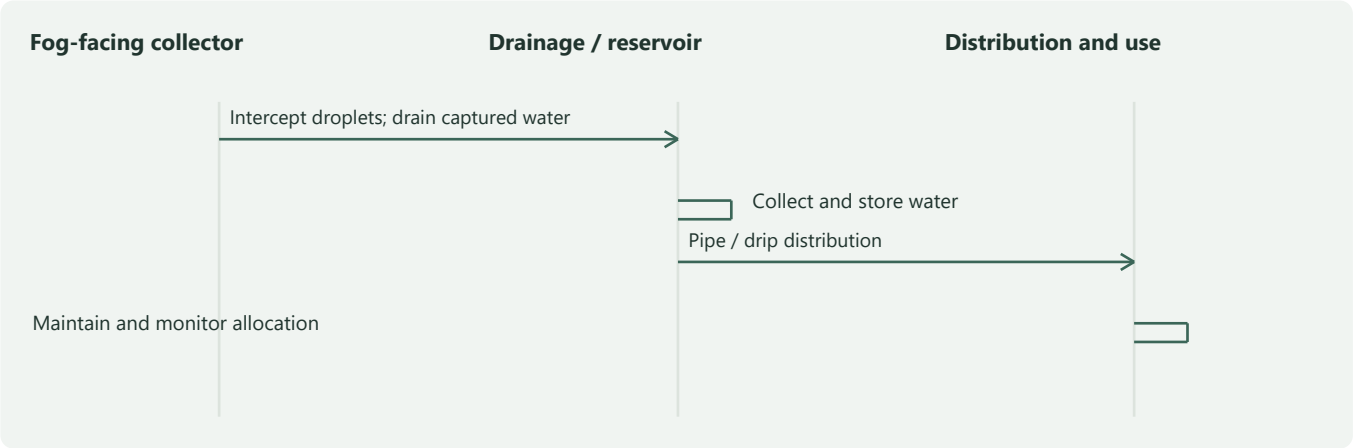
Farmers describe dry wells or tanks, unaffordable alternatives and unreliable utilities. Some report local winter fog and existing bamboo cultivation. These observations motivate a locally supplied water-collection concept, but recollections of fog do not quantify recoverable water. The design must separately establish climate suitability, production capability and social acceptance.

Source activity	Role
Village questionnaires	Needs, existing practices and conditional interest
NGO / expert interviews	Implementation and livelihood context
Model / solution diagram	Explain the proposed resource flows

Source / interpretation: Droplets of Hope SBV-12-SO.SCIENCE (2).pdf, pp1,4,19-30. Cover and approval list Jahnvi Julka, Namya Doongarwal, Arham Sethi and Yajat Khanna.

Bamboo fog collection and water flow

The concept uses bamboo-fibre collection surfaces, drainage and storage to connect a fog-exposed location with downstream agricultural use. Its source illustrations show the proposed mechanism and landscape arrangement. They should be read as design explanations, with performance requiring site-specific measurement.



Mechanism and model

The portfolio describes hydrophilic and hydrophobic bamboo-fibre regions directing collected moisture toward tanks, with pipes transporting water downhill. Its model uses a 46.5 by30 inch inclined board to depict hilly terrain. Those dimensions describe the explanatory model, not the scale of an installed collector or a tested relationship to real hillside altitude.

Checking the yield calculation

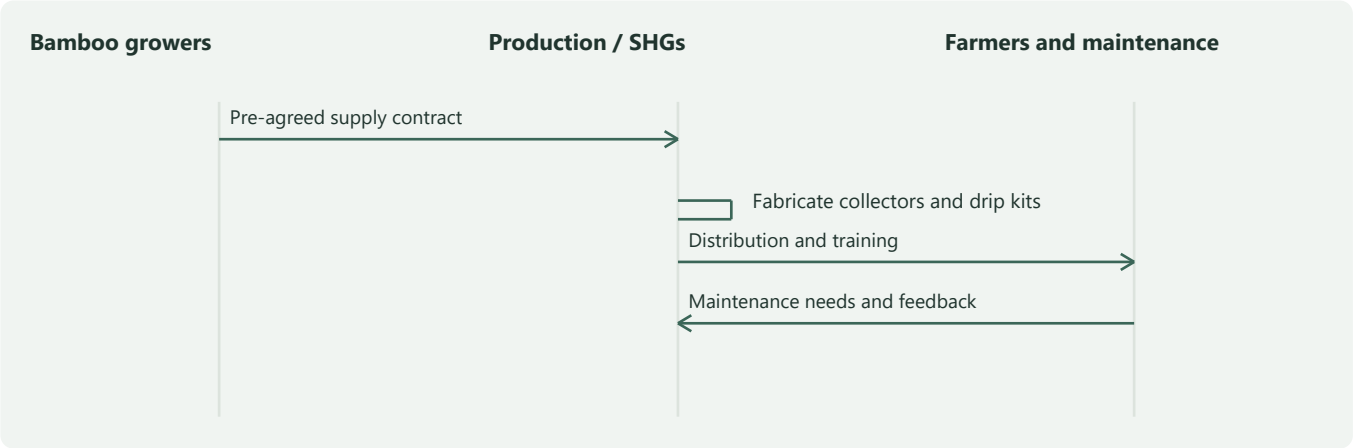
Page32 assumes 8 litres per square metre per day and multiplies by56.6 square metres to obtain452.8 litres per day under favourable conditions. The same page gives net dimensions4 by3 metres, whose projected area is12 square metres; it does not transparently derive56.6. Effective fibre surface and collector frontal area cannot be interchanged without a defined model. Treat this as an unreconciled estimate, not a measured yield.

Quantity	Evidence interpretation
4m x3m	Specified net dimensions;12m2 projected area
56.6m2 and8L/m2/day	Source assumptions requiring reconciliation
452.8L/day	Calculated scenario, not observed field collection

Source / interpretation: Krishi Connect source pp31-32,37-40. Reuse landscape/model diagrams on p37 and flow explanation p38, retaining their proposed status.

Bamboo supply and community delivery

The design links agricultural raw material, local fabrication and water-system maintenance. This social supply chain is central: technical equipment alone does not establish an affordable or maintained service. The portfolio proposes connecting growers, tribal producers, NGOs, existing self-help groups and panchayats.



Conditional willingness

Page34 reports interactions with20 farmers, of whom17 agreed to grow bamboo if offered market price under predetermined contract terms. This is conditional stated willingness in a small interviewed group. It is not evidence of planted acreage, signed contracts, delivered bamboo or sustained adoption, and should not be generalized as a population acceptance rate.

Proposed delivery structure

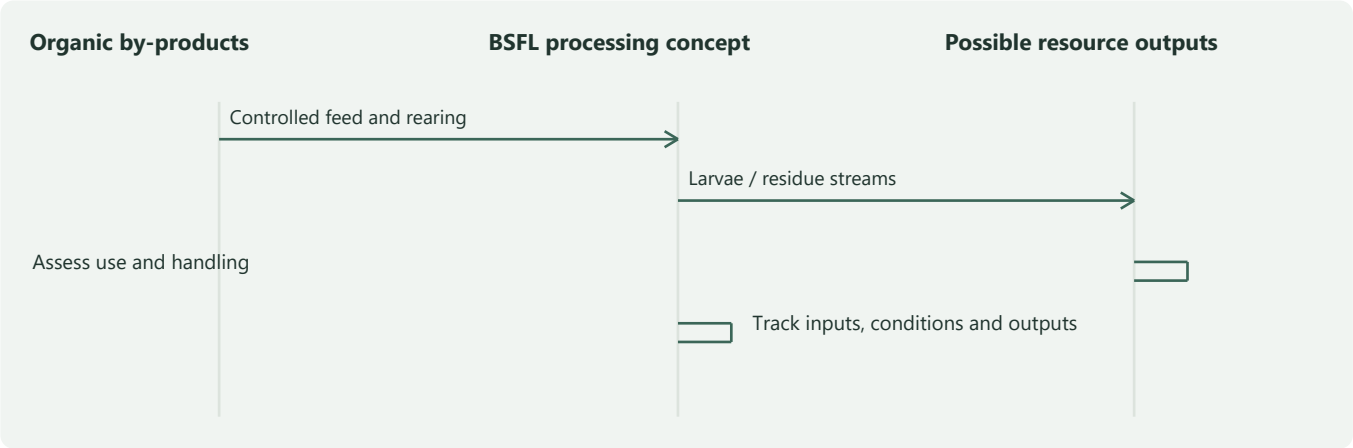
The source envisages NGO-mediated employment for tribal craftspeople, provision of drip kits to existing SHGs and panchayat oversight of distribution. It names bamboo poles/fibres and hemp or jute/twine as inputs. The future-tense implementation section indicates intended partnerships and roles; it does not establish that the named organizations committed resources or that the programme launched.

Actor	Proposed responsibility
Growers	Supply bamboo under agreed terms
NGOs / producers / SHGs	Fabrication, capacity and maintenance
Panchayat / users	Distribution governance and feedback

Source / interpretation: Krishi Connect source pp20-24,30,33-34 and39. The reported20/17 group concerns willingness to supply bamboo, not measured beneficiaries.

Circular resource use and outreach

Alongside water collection and drip delivery, the portfolio explores black soldier fly larvae for organic-waste processing and livelihood support. The work also documents public awareness activity. These strands share a resource-management theme while requiring different evidence to assess their effects.



BSFL as a researched option

The source describes larvae consuming organic material and producing possible feed and fertilizer-related outputs. It gives environmental ranges and broad resource-saving claims drawn from background research. No controlled rearing trial, waste-conversion measurement or verified water-saving outcome is documented here. Those quantities should therefore stay out of measured project results.

Documented communication

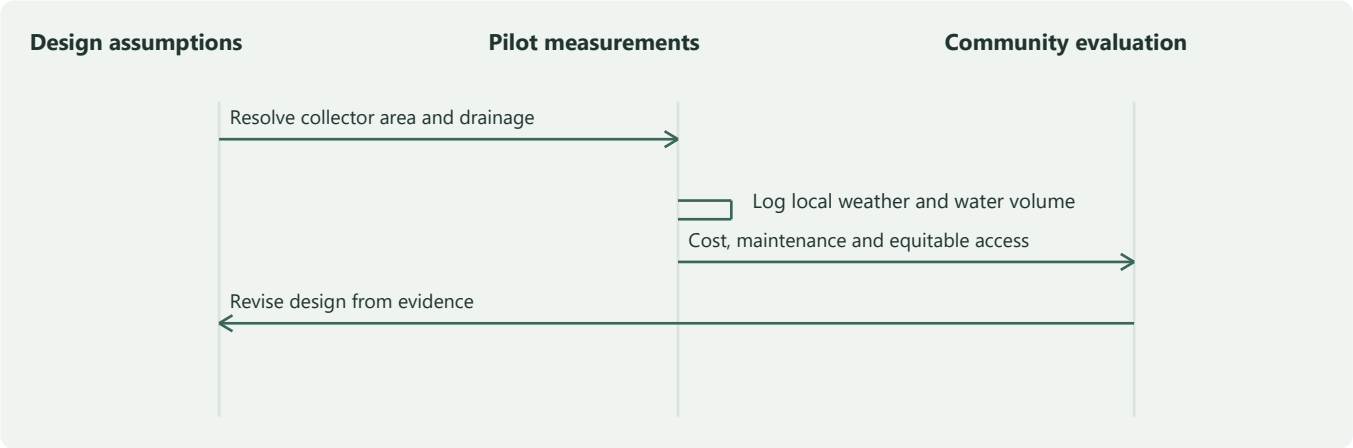
Page29 records a street play at Jhalaria village and Chhappan Dukaan to raise awareness of tribal welfare and water scarcity, accompanied by photographs. This supports an outreach activity claim. Attendance, changes in awareness and subsequent behaviour are not quantified, so impact should not be inferred from the presence of an event or photographs alone.

Output	Supported account
BSFL discussion	Researched complementary concept
Street play and photographs	Documented outreach activity
Integrated model	Explains proposed water and livelihood connections

Source / interpretation: Krishi Connect source pp25-29,35-38 and42. Background biological claims are not independently verified technical recommendations.

Validation priorities and project learning

Krishi Connect demonstrates an integrated research-and-design approach: listen to community constraints, study potential materials and mechanisms, and propose a governance structure for delivery. Its strongest current outputs are the questionnaires, recorded interactions, model and solution architecture.



Technical validation

A pilot should define frontal area, mesh material, porosity and orientation, then log collected volume with fog, wind and rainfall conditions. Rainfall contributions and storage losses need separate accounting. Any planned drinking-water use also needs appropriate water-quality assessment. These are validation requirements derived from the proposed uses, not completed tests.

Social and operational validation

A delivery trial should distinguish expressions of interest from actual contracts, employment and functioning systems. Recording training, payments, repair frequency, distribution rules and user feedback would test the proposed organization. The source's title and model can be preserved while clearly presenting projected performance as a hypothesis and community-research findings as the documented foundation.

Evidence to collect	Question answered
Water / climate log	How much usable water is captured under local conditions?
Cost and repair record	Can the system remain affordable and functional?
Participation / contract record	Which proposed roles become sustained activity?

Source / interpretation: Krishi Connect source pp19-24,30-40 and42. Proposed next measurements extend the documented design; they are not claimed as accomplished outcomes.

Evidence & source

Implementation, supporting records, and the boundary of the claim

Area	Evidence	Reference
Primary record	Supplied dissertation	Original PDF linked unchanged
Methods	Questionnaires, interviews and explanatory models	Companion source-page references
Attribution	Team research; Arham Sethi	Original team credits retained

Primary references

File / record	Path / reference
Original dissertation	documents/krishi-connect-dissertation.pdf

Original dissertation retained unchanged. Project name and year follow the user's clarification; the document internally uses the title Droplets of Hope.

Evidence boundary

Team community research and proposed intervention design. Survey responses, expert projections, models and observed awareness activities are distinguished from measured long-term agricultural, water or income outcomes.

Technical note

The companion explains sampling, consultation, proposed processes and evidence limits. Quantities in projections are not automatically treated as delivered outcomes.